

Table 1: Experiment 8: SafeDreamer vs MACPO on MAMuJoCo (continuous actions).

		Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$
2x3HalfCheetah	5.0	MACPO	Paper		10M	2250.0	40.0
2x3HalfCheetah	5.0	MACPO	Run		10M	1316.8 ± 115.7	5.0 ± 1.7
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		100k	8.5	0.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		500k	1410.6	0.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		700k	1455.9	15.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		800k	1450.3	22.2
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		900k	1685.9	37.8
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5)		1M	1519.2	35.8
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		100k	393.4 ± 466.5	0.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		500k	1554.6 ± 631.6	0.8 ± 1.3
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		700k	1801.6 ± 696.3	4.6 ± 4.1
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		800k	1931.8 ± 820.7	127.0 ± 208.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		900k	1744.9 ± 724.4	66.8 ± 100.4
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-4)		1M	1480.8 ± 97.6	4.6 ± 2.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		100k	445.1 ± 454.1	0.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		500k	717.3 ± 963.0	15.4 ± 21.4
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		700k	728.8 ± 1041.0	0.6 ± 0.5
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		800k	838.8 ± 1160.6	3.2 ± 3.0
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		900k	722.3 ± 439.0	4.8 ± 7.7
2x3HalfCheetah	5.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1166.0 ± 512.6	11.1 ± 18.9
2x3HalfCheetah	25.0	SafePO	Paper		10M	1637.3	52.1
2x3HalfCheetah	25.0	MACPO	Run		10M	1374.3 ± 85.7	22.0 ± 4.2
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		100k	1136.2 ± 564.6	30.1 ± 30.2
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		500k	2000.5 ± 120.6	5.4 ± 6.9
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		700k	2587.8 ± 181.4	22.3 ± 22.2
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		800k	2545.2 ± 212.5	37.6 ± 9.4
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		900k	2528.5 ± 162.9	3.9 ± 3.7
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5)		1M	2527.2 ± 228.6	16.6 ± 2.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		100k	-60.1 ± 39.0	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		500k	743.2 ± 408.4	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		700k	969.0 ± 587.9	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		800k	1019.7 ± 396.6	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		900k	1070.2 ± 308.5	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-4)		1M	1149.1 ± 312.7	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		100k	409.9 ± 197.0	0.0
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		500k	960.4 ± 942.3	45.2 ± 52.7
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1581.8 ± 590.2	41.8 ± 40.3
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1120.2 ± 467.7	18.2 ± 17.5
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		900k	824.5 ± 464.8	2.9 ± 1.8
2x3HalfCheetah	25.0	SafeDreamer	(lr=1e-5, nocomm)		1M	981.5 ± 1303.1	64.4 ± 103.8
2x3HalfCheetah	25.0	MAPPO-Lag	Run		10M	1141.3 ± 35.9	0.0
2x4Ant	0.2	MACPO	Paper		10M	900.0	15.0
2x4Ant	0.2	MACPO	Run		10M	859.2 ± 32.7	1.2 ± 0.7
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		100k	894.0 ± 51.4	0.0
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		500k	1410.2 ± 394.7	2.0 ± 1.8
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		700k	1566.0 ± 857.2	18.0 ± 11.8
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		800k	1626.6 ± 158.9	43.5 ± 8.2
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		900k	1902.4 ± 359.2	43.9 ± 29.7
2x4Ant	0.2	SafeDreamer	(lr=1e-5)		1M	2267.8 ± 134.1	24.3 ± 5.6
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		100k	903.4 ± 27.3	0.0
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		500k	1187.2 ± 137.5	0.0
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		700k	1184.0 ± 499.9	5.1 ± 5.0
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		800k	1373.5 ± 727.7	2.0 ± 2.8

Table 1 – continued

		Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$	
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		900k	1494.8 $\pm$ 831.1		3.7 $\pm$ 6.0
2x4Ant	0.2	SafeDreamer	(lr=1e-4)		1M	1618.4 $\pm$ 640.8		4.1 $\pm$ 5.8
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		100k	887.0 $\pm$ 13.3		0.0
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		500k	815.9 $\pm$ 181.1		8.6 $\pm$ 14.8
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		700k	826.6 $\pm$ 252.1		3.9 $\pm$ 6.8
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		800k	1102.1 $\pm$ 223.0		2.6 $\pm$ 4.5
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		900k	1126.3 $\pm$ 281.9		1.7 $\pm$ 3.0
2x4Ant	0.2	SafeDreamer	(lr=1e-5, nocomm)		1M	1191.6 $\pm$ 444.6		0.9 $\pm$ 1.2
2x4Ant	25.0	SafePO	Paper		10M	1099.2		3.4
2x4Ant	25.0	MACPO	Run		10M	798.1 $\pm$ 109.4		10.5 $\pm$ 1.7
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		100k	836.2 $\pm$ 93.2		0.0
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		500k	1086.0 $\pm$ 412.1		6.8 $\pm$ 11.5
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		700k	1613.5 $\pm$ 553.1		53.1 $\pm$ 47.1
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		800k	1794.4 $\pm$ 672.1		82.9 $\pm$ 72.0
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		900k	1838.3 $\pm$ 583.0		32.7 $\pm$ 33.7
2x4Ant	25.0	SafeDreamer	(lr=1e-5)		1M	1835.8 $\pm$ 678.3		17.4 $\pm$ 15.1
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		100k	917.0 $\pm$ 58.0		0.0
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		500k	1207.0 $\pm$ 175.0		0.1 $\pm$ 0.2
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		700k	1156.7 $\pm$ 477.8		1.4 $\pm$ 1.2
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		800k	1221.1 $\pm$ 478.7		2.0 $\pm$ 0.6
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		900k	1323.1 $\pm$ 629.2		7.0 $\pm$ 7.2
2x4Ant	25.0	SafeDreamer	(lr=1e-4)		1M	1552.4 $\pm$ 475.2		22.6 $\pm$ 18.3
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		100k	918.4 $\pm$ 25.2		0.0
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		500k	1083.4 $\pm$ 114.7		0.0
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1216.5 $\pm$ 243.5		1.0 $\pm$ 1.1
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1221.0 $\pm$ 60.1		1.3 $\pm$ 1.3
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		900k	1505.5 $\pm$ 345.8		4.7 $\pm$ 7.6
2x4Ant	25.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1491.0 $\pm$ 79.8		19.4 $\pm$ 30.3
2x4Ant	25.0	MAPPO-Lag	Run		10M	1579.6 $\pm$ 185.0		1.4 $\pm$ 0.3
4x2Ant	1.0	MACPO	Paper		10M	650.0		15.0
4x2Ant	1.0	MACPO	Run		10M	1097.6 $\pm$ 97.4		3.4 $\pm$ 1.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		100k	916.3 $\pm$ 15.3		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		500k	965.2 $\pm$ 34.3		0.3 $\pm$ 0.6
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		700k	886.0 $\pm$ 110.8		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		800k	1134.0 $\pm$ 170.2		0.1 $\pm$ 0.1
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		900k	1140.6 $\pm$ 33.1		0.5 $\pm$ 0.8
4x2Ant	1.0	SafeDreamer	(lr=1e-5)		1M	1241.0 $\pm$ 565.5		2.5 $\pm$ 3.1
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		100k	960.3 $\pm$ 33.3		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		500k	1069.5 $\pm$ 106.2		0.1 $\pm$ 0.1
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		700k	1432.8 $\pm$ 404.7		4.2 $\pm$ 7.3
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		800k	1378.4 $\pm$ 295.2		7.7 $\pm$ 13.4
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		900k	1595.7 $\pm$ 410.7		2.5 $\pm$ 4.0
4x2Ant	1.0	SafeDreamer	(lr=1e-4)		1M	1558.8 $\pm$ 359.0		4.2 $\pm$ 7.3
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		100k	901.8 $\pm$ 77.0		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		500k	850.9 $\pm$ 101.8		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		700k	1105.4 $\pm$ 193.7		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		800k	1104.5 $\pm$ 140.4		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		900k	1159.3 $\pm$ 189.8		0.0
4x2Ant	1.0	SafeDreamer	(lr=1e-5, nocomm)		1M	1032.4 $\pm$ 57.2		0.0
4x2Ant	25.0	SafePO	Paper		10M	815.8		0.0
4x2Ant	25.0	MACPO	Run		10M	1287.3 $\pm$ 148.3		23.6 $\pm$ 6.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		100k	902.4 $\pm$ 49.0		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		500k	997.3 $\pm$ 83.4		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		700k	1192.7 $\pm$ 86.6		0.0
4x2Ant	25.0	SafeDreamer	(lr=1e-5)		800k	1265.9 $\pm$ 101.8		0.0

Table 1 – continued

	Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$	
4x2Ant	25.0		SafeDreamer (lr=1e-5)		900k	1406.8 ± 172.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)		1M	1574.5 ± 304.2	0.5 ± 0.8
4x2Ant	25.0		SafeDreamer (lr=1e-4)		100k	918.7 ± 12.6	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)		500k	1025.5 ± 69.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)		700k	1168.3 ± 497.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)		800k	1221.5 ± 220.3	0.1 ± 0.2
4x2Ant	25.0		SafeDreamer (lr=1e-4)		900k	1260.2 ± 302.6	0.7 ± 1.2
4x2Ant	25.0		SafeDreamer (lr=1e-4)		1M	1504.2 ± 520.0	2.9 ± 4.8
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		100k	939.3 ± 55.0	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		500k	1164.9 ± 175.7	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		700k	1301.5 ± 286.0	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		800k	1335.0 ± 278.6	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		900k	1515.5 ± 253.9	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5, nocomm)		1M	1656.8 ± 161.8	0.5 ± 0.9
4x2Ant	25.0		MAPPO-Lag Run		10M	1865.5 ± 41.9	0.1 ± 0.2